

CS4820 Prelim 1 Solution

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Algorithm 1 Greedy Sequence Alignment

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 $\ell \leftarrow 1$ 
 $h \leftarrow 1$ 
 $A \leftarrow []$ 
while  $\ell \leq k$  and  $h \leq n$  do
    if  $L[\ell] = H[h]$  then                                      $\triangleright$  There exists an alignment between  $L$  and  $H$ 
         $A.append(h)$ 
         $\ell \leftarrow \ell + 1$ 
    end if
     $h \leftarrow h + 1$ 
end while
if  $len(A) = k$  then                                            $\triangleright L$  is fully aligned
    return  $A$ 
else
    return  $\perp$ 
end if
```

Correctness can also be proven without greedy stays ahead. Consider the following two lemmas, the second of which utilizes an exchange argument:

Lemma 0.1. *If our algorithm returns an array A , then A is a non-crossing alignment.*

Proof. This can be proven by induction or by arguing in terms of loop invariants (really, both these approaches are the same). First, note that the variable ℓ is nondecreasing throughout the course of the loop. Furthermore, any time a value of ℓ is appended to A , we update $\ell \leftarrow \ell + 1$. Thus, for indices $i < j \leq |A|$, we have $A[i] < A[j]$. The fact that $H_{A[\ell]} = L_\ell$ for all $\ell \in [|A|]$, follows from the fact that we only append h to A when the condition $L_\ell = H_h$ is true.

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□

Lemma 0.2. *If an alignment exists, our algorithm returns a valid alignment.*

Proof. Suppose a sequence alignment O exists. Let $s = |A|$. Consider the first index at which A and O differ. We consider two possibilities:

- $\forall i \in [s], A[i] = O[i]$ but $s < |O| = k$.

This implies that our algorithm failed to find an instance of character $L[s+1]$ after the $A[s]$ -th element in the sequence H . However, by assumption $H_{O[s+1]} = L_{s+1}$ and $O[s+1] > O[s] = A[s]$. After the algorithm appends the last element of A , the variable ℓ is equal to $s+1 \leq |O| = k$ for the remainder of the algorithm (as ℓ is only updated when we append an element to A). Thus, we keep iterating until $h = n$. Furthermore, note that on the iteration where $h = O[s+1] \leq n$, our algorithm performs the check $L_{s+1} = H_{O[s+1]}$. By optimality of O , this check passes and so our algorithm must have appended $O[s+1]$ to A . A contradiction! $\nexists$

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- $\exists j \in [\ell]$ such that $\forall i < j, A[i] = O[i]$ but $A[j] \neq O[j]$.

We claim that we can just replace $O[j]$ in O with $A[j]$ to obtain a valid sequence alignment O' that agrees with A on the first j indices. For this, it suffices to show that $A[j] \leq O[j]$ as we already know $H_{A[j]} = L_j = H_{O[j]}$.

By the fact that our algorithm only appends to A when $L[\ell] = H[h]$, it must be the case that for $h \in \{A[\ell - 1] + 1, \dots, A[\ell] - 1\}$, we have $L[\ell] \neq H[h]$. Applying this to $\ell = j$ and using the fact that $O[j] > O[j - 1] = A[j - 1]$, we can then conclude that $O[j] \geq A[j]$.

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